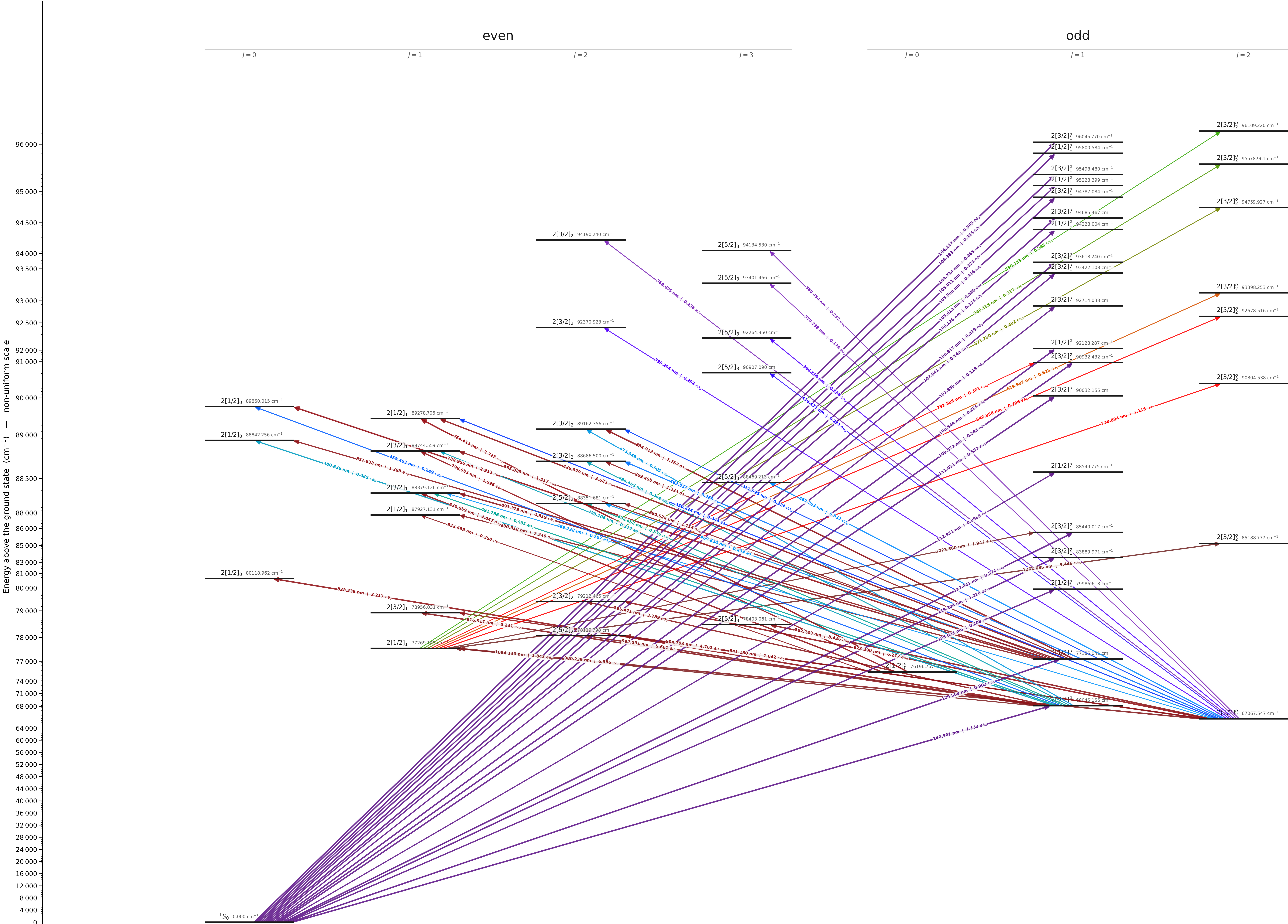


Xe (Xe I). Strongest classified transitions below 2 μm (NIST): 51 levels, 70 lines.

Xe+ ionisation limit 97833.79 cm⁻¹ = 12.12984 eV (102.214 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U e r U (ea₀)	A (s⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
1S₀ - 2[3/2]₁	104.1170	-	2879.3798	0.363	7.900e+07	-	-	-
1S₀ - 2[1/2]₁	104.3835	-	2872.0293	0.315	5.900e+07	-	-	-
1S₀ - 2[3/2]₁	104.7137	-	2862.9724	0.465	1.270e+08	-	-	-
1S₀ - 2[3/2]₁	105.4996	-	2841.6453	0.316	5.750e+07	-	-	-
1S₀ - 2[3/2]₁	105.6128	-	2838.5989	0.580	1.930e+08	-	-	-
1S₀ - 2[3/2]₁	106.8168	-	2806.6042	0.819	3.720e+08	-	-	-
1S₀ - 2[1/2]₁	108.5443	-	2761.9366	0.285	4.280e+07	-	-	-
1S₀ - 2[3/2]₁	109.9718	-	2726.0857	0.283	4.080e+07	-	-	-
1S₀ - 2[3/2]₁	111.0714	-	2699.0961	0.552	1.500e+08	-	-	-
1S₀ - 2[3/2]₁	117.0412	-	2561.4273	0.574	1.390e+08	-	-	-
1S₀ - 2[3/2]₁	119.2038	-	2514.9581	1.220	5.930e+08	-	-	-
1S₀ - 2[1/2]₁	129.5588	-	2313.9493	0.903	2.530e+08	-	-	-
1S₀ - 2[3/2]₁	146.9612	-	2039.9425	1.133	2.730e+08	-	-	-
2[1/2]₁ - 2[1/2]₀	788.9563	788.7393	379.9862	2.913	3.500e+07	-	-	-
2[3/2]₁ - 2[1/2]₀	828.2392	828.0116	361.9636	3.217	3.690e+07	-	-	-
2[1/2]₁ - 2[3/2]₂	834.9117	834.6823	359.0709	7.767	4.200e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).